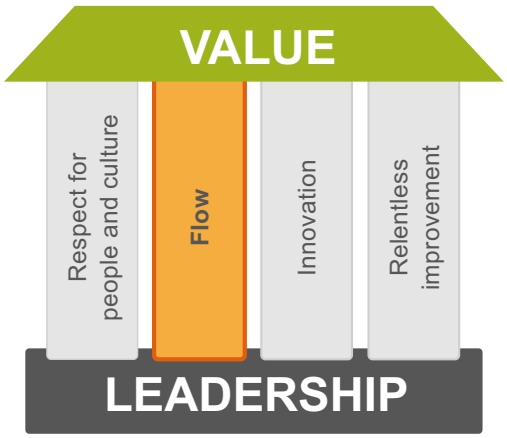
Flow

- Optimize sustainable value delivery
- Build in quality
- Understand, exploit, and manage variability
- Move from projects to products



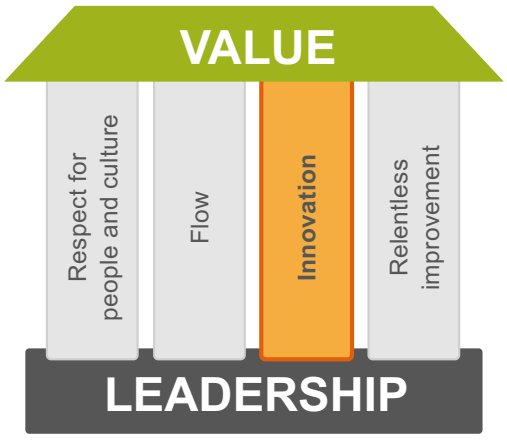
Operating a product development process near full utilization is an economic disaster.

—Don Reinertsen

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Innovation

- Innovative people
- Provide time and space for innovation
- Go see
- Experimentation and feedback
- Innovation riptides
- Pivot without mercy or guilt



Innovation comes from the producer.

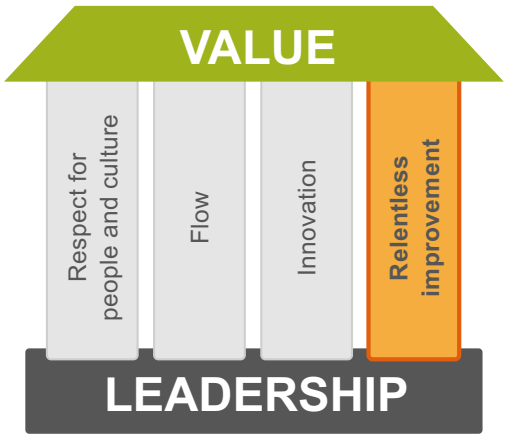
—W. Edwards Deming

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Relentless improvement

- A constant sense of danger
- Optimize the whole
- Problem-solving culture
- Base improvements on facts
- Reflect at key Milestones

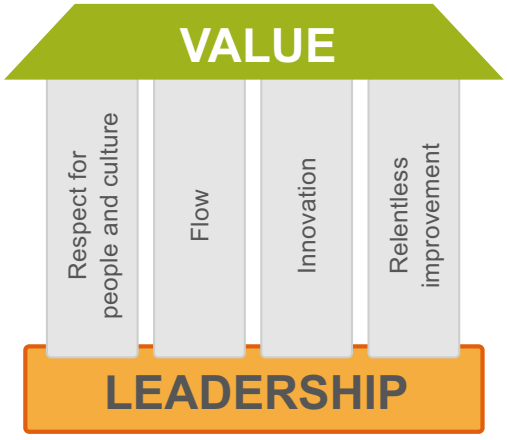


Those who adapt the fastest win.

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Leadership

- Lead by example
- Adopt a growth mindset
- Exemplify the values and principles of Lean-Agile and SAFe
- Develop people
- Lead the change
- Foster psychological safety



People are already doing their best; the problems are with the system. Only management can change the system.

—W. Edwards Deming

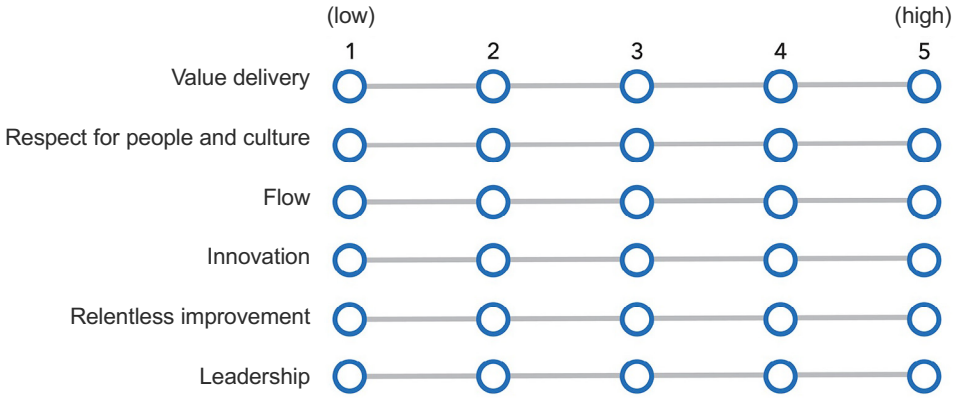
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Activity: Assessing a Lean mindset

Duration: 5 min

- **Step 1:** Assess where your team stands in embracing a Lean mindset.
- **Step 2:** Discuss the results of the self-assessment. Do you have similar low or high scores?



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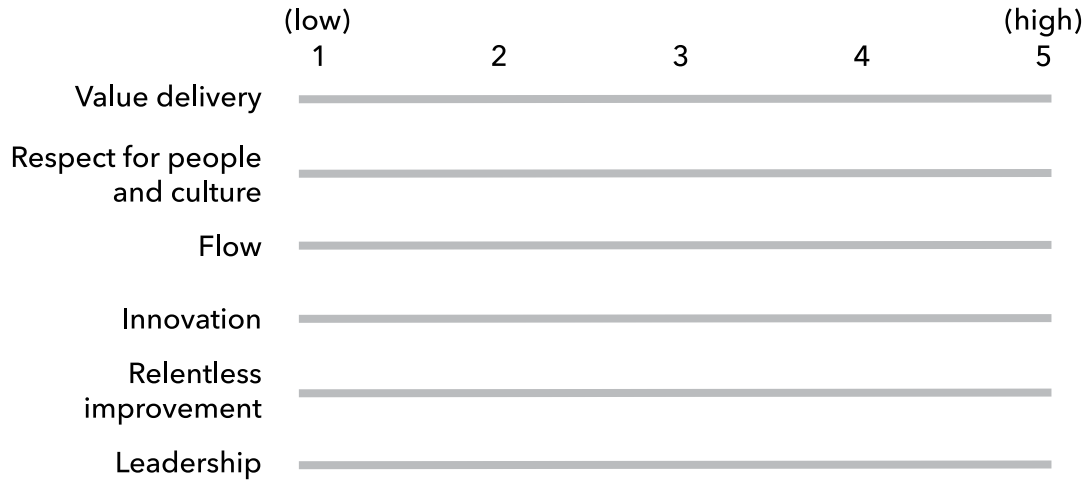
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Lean Mindset Self-Assessment

Step 1: Assess where your team stands in embracing a Lean mindset.

Step 2: Discuss the results of the self-assessment. Do you have similar low or high scores?



Notes

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The Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over processes and tools

Working software over comprehensive documentation

Customer collaboration over contract negotiation

Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

https://agilemanifesto.org/

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The Agile Manifesto Principles

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference for the shorter timescale.
4. Business people and developers must work together daily throughout the project.

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The Agile Manifesto Principles

5. Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
7. Working software is the primary measure of progress.
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

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The Agile Manifesto Principles

9. Continuous attention to technical excellence and good design enhances agility.
10. Simplicity—the art of maximizing the amount of work not done—is *essential.*
11. The best architectures, requirements, and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

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Activity: Agile Manifesto principles

Prepare: 5 min

Share: 3 min

- **Step 1:** Review the principles behind the Agile Manifesto.
- **Step 2:** As a group, select one or more principles to discuss.
- **Step 3:** In your group, discuss how these principles apply in your context.
- **Step 4:** Provide an example of how you would apply one of the principles to your context and share with class.

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1.4 Lean and Agile at scale with the SAFe Principles

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SAFe Lean-Agile Principles

#1 Take an economic view

#2 Apply systems thinking

#3 Assume variability; preserve options

#4 Build incrementally with fast, integrated learning cycles

#5 Base milestones on objective evaluation of working systems

#6 Visualize and limit WIP, reduce batch sizes, and manage queue lengths

#7Apply cadence, synchronize with cross-domain planning

#8 Unlock the intrinsic motivation of knowledge workers

#9 Decentralize decision-making

#10 Organize around value

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Video: Building Incrementally: Economic Advantage

Duration 3 min

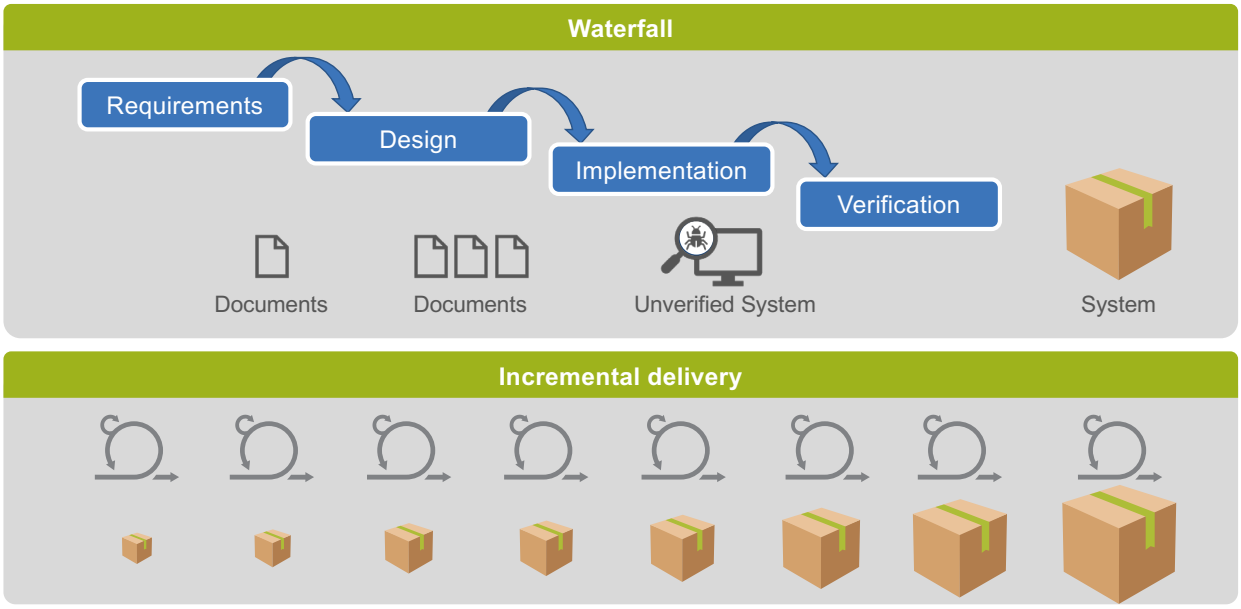


https://bit.ly/Video-BuildingIncrementally

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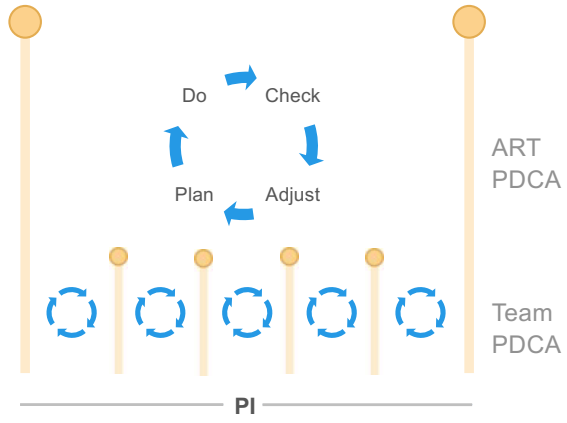
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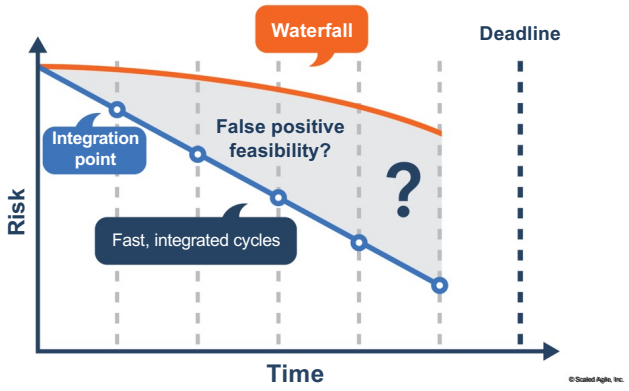
Agile economics: Deliver early and often



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Use Iterations and Program Increments to learn quickly



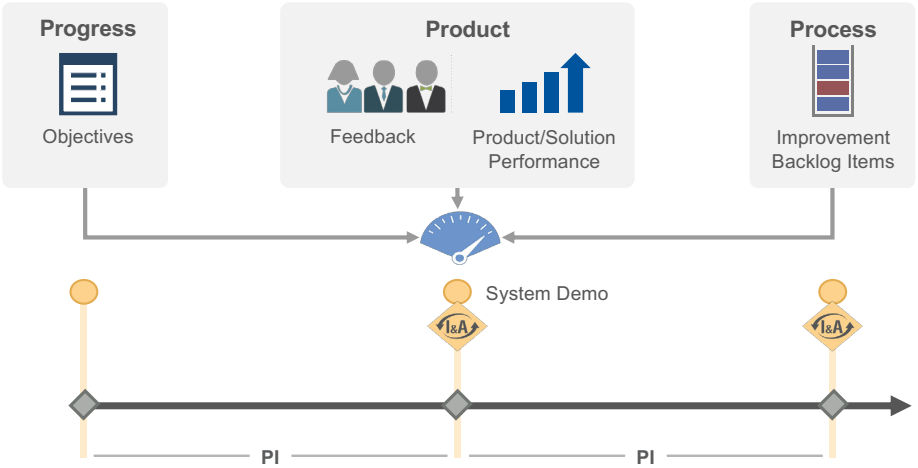


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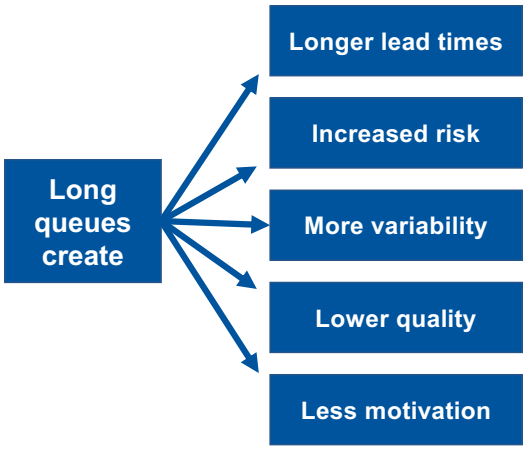
Base milestones on objective evaluation of working systems

Program Increment (PI) System Demos are orchestrated to deliver objective progress, product, and process Metrics.



Visualize and limit WIP, reduce batch size, and manage queue lengths

Long queues: All bad



Principles of Product Development Flow, Don Reinertsen

Little's Law

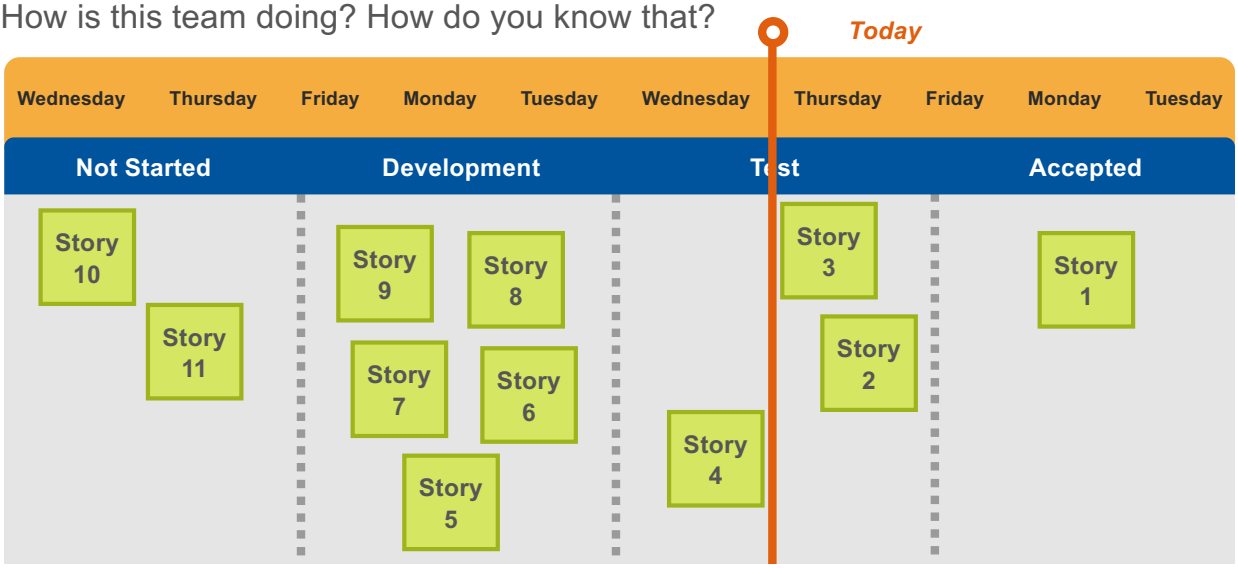
$$W_q = \frac{L_q}{\lambda}$$

Average wait time = average queue length divided by average processing rate

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Visualize and limit work in progress

How is this team doing? How do you know that?



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Activity: Experience a large batch size

Duration 5 min

- **Step 1:** Create groups of five people with 10 coins per group. Designate one person as the timekeeper. The remaining four people will be processing the coins.
- **Step 2:** Person by person process each coin.
- **Step 3:** Pass all coins at the same time to the next person, who repeats step two until all four people are done
- **Step 4:** The timekeeper stops the timer and records the total time



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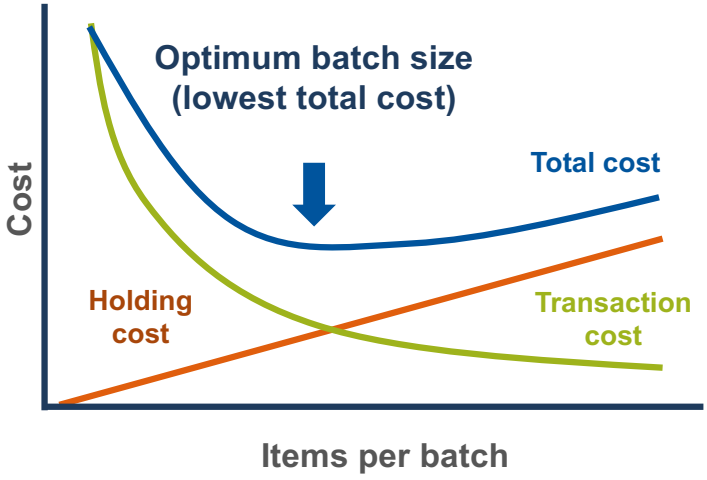
Activity: Experience a small batch size

Duration: 5 min

- **Step 1:** Ensure that the timekeeper is ready to start the timer
- **Step 2:** This time, each person processes one coin at a time and immediately passes each coin to the next person
- **Step 3:** The timekeeper will stop the timer when the last person flips the last coin and records the result



Reduce batch size for higher predictability

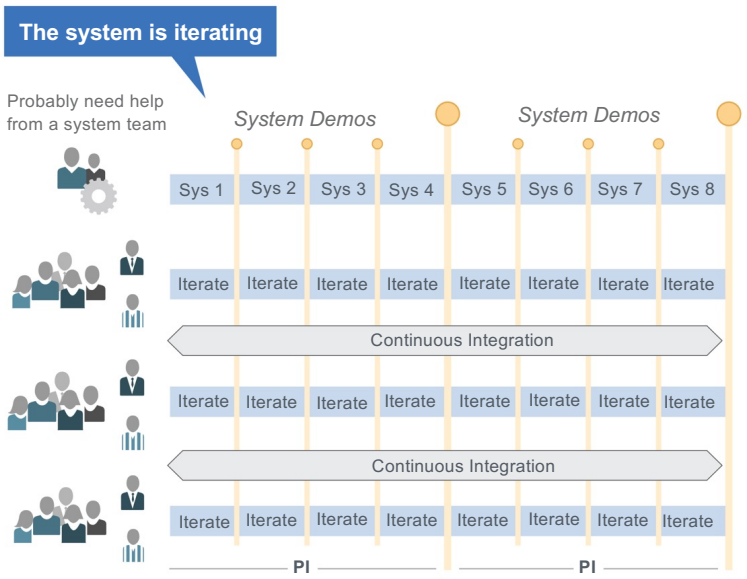


Principles of Product Development Flow, Don Reinertsen

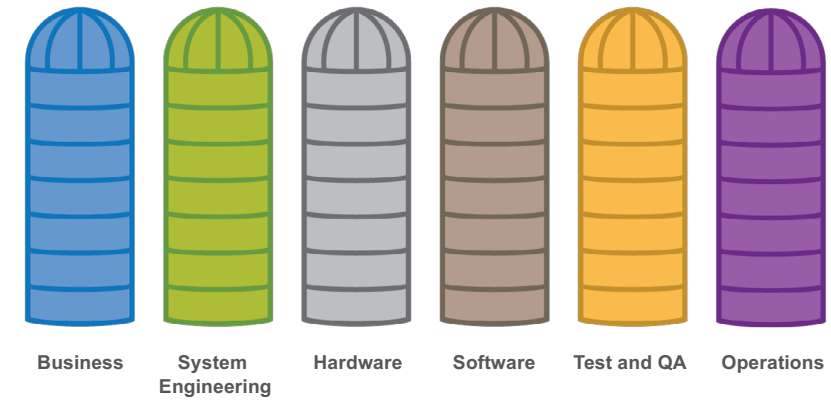
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Apply cadence, synchronize with cross-domain planning

Cadence-based planning limits variability.



Value doesn't follow silos



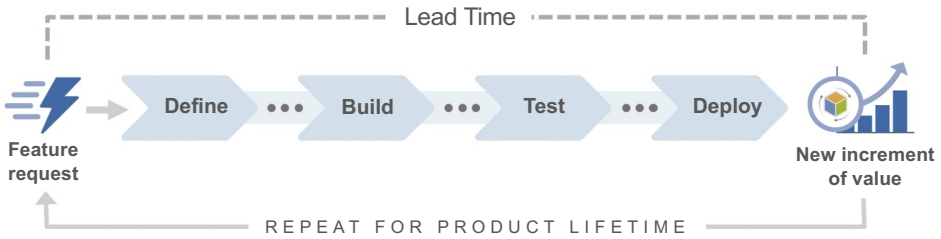
- Value delivery is inhibited by handoffs and delays
- Political boundaries can prevent cooperation
- Silos encourage geographic distribution of functions
- Communication across silos is difficult

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Instead, organize around Development Value Streams

The aim of development is in fact the creation of profitable operational value streams.
—Allen C. Ward

- Includes activities from recognizing an opportunity through release and validation
- Contains the steps, the flow of information and material, and the people who develop the Solutions used by the Operational Value Streams



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1.5 Scrum, Kanban, and Quality Practices

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From traditional development to Agile

Instead of a large group…



…working on all of the requirements…



…and integrating and delivering value toward the end of development,

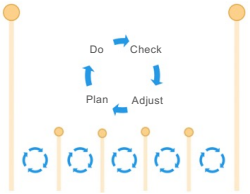




Have small teams working together as a program…



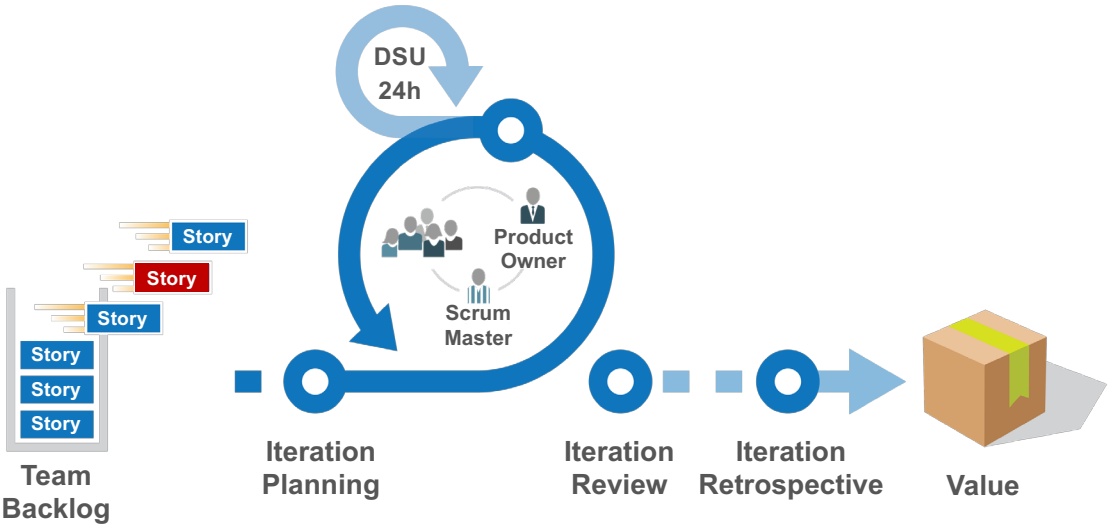
…working on small batches of requirements…



…and delivering value in short timeboxes with frequent integration and improvement cycles.

Agile for teams: Scrum

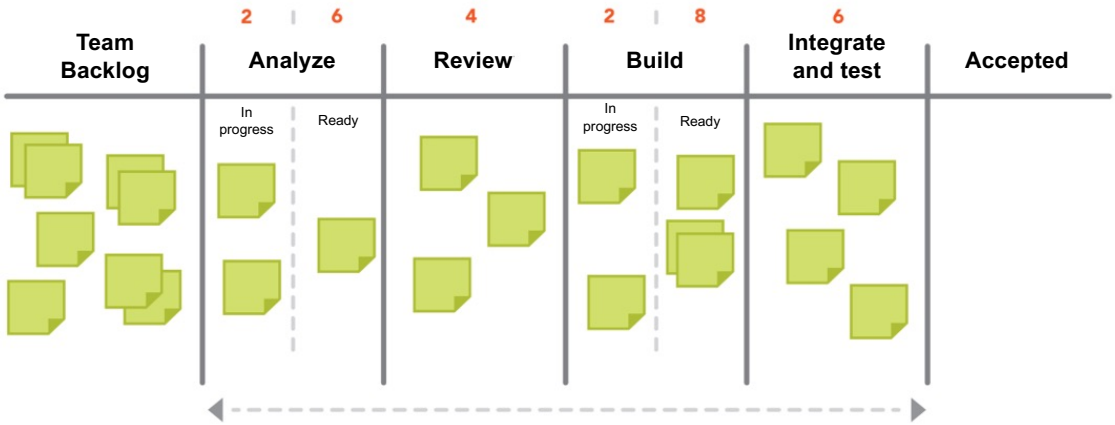
Scrum is built on transparency, inspection, adaptation, and short learning cycles.



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Agile for teams: Kanban

Visualize work flow. Limit work in process. Improve flow.

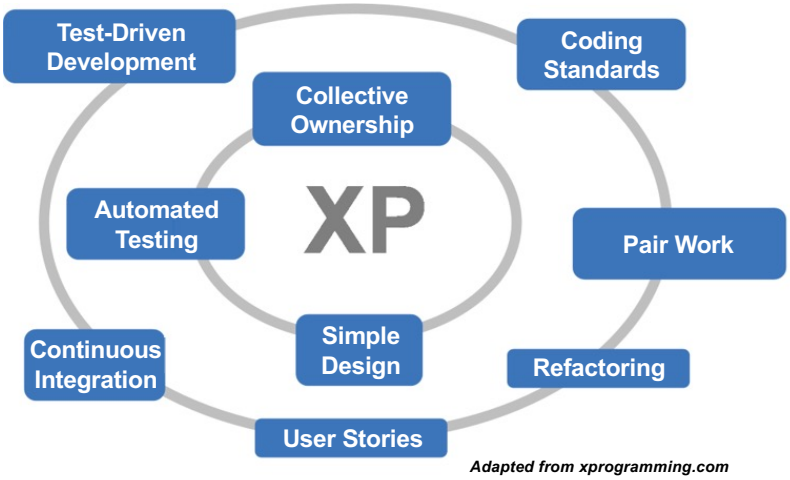


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Quality practices provides the basis for Technical Agility

Examples of quality practices inspired by eXtreme Programming (XP)



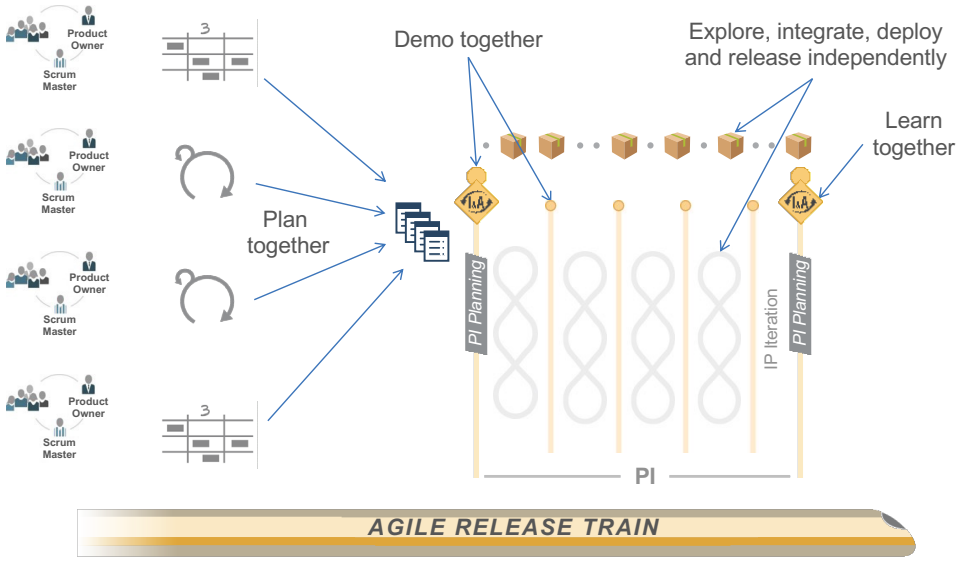
Adapted from xprogramming.com

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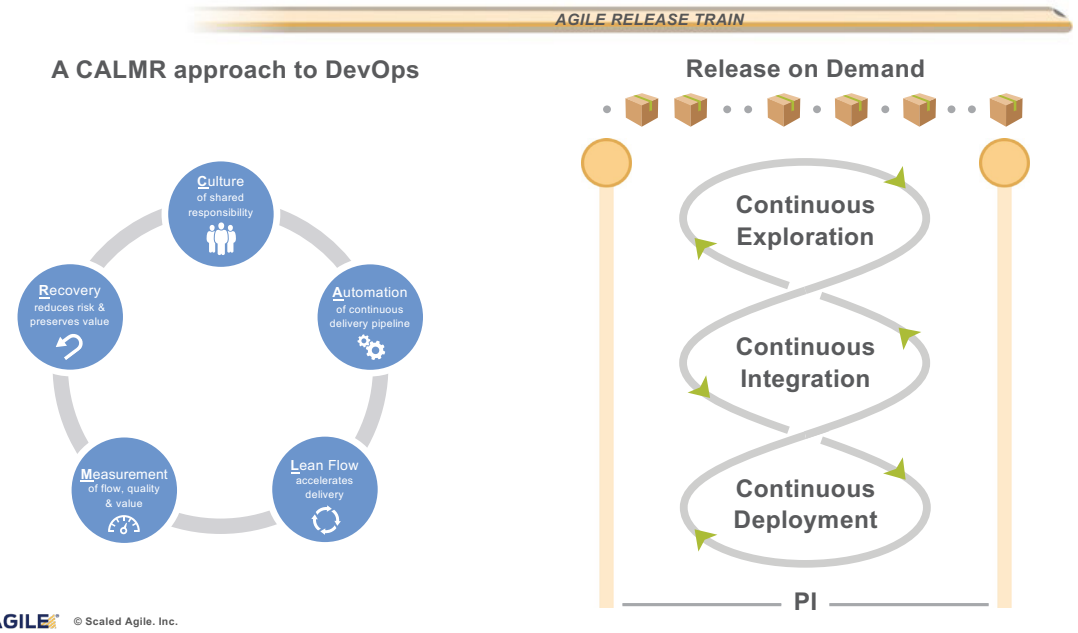
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Teams in SAFe are part of an Agile Release Train



The ART and teams continuously deliver value



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Action Plan: Introducing SAFe

Prepare 5 min

Share 3 min

- **Step 1:** As a team, brainstorm one to three actions you could take to improve in any areas related to this lesson
- **Step 2:** Individually write down at least one improvement item
- **Step 3:** Share one item you discussed as a team and one item you individually wrote in your Action Plan



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Introducing SAFe

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Lesson review

In this lesson you:

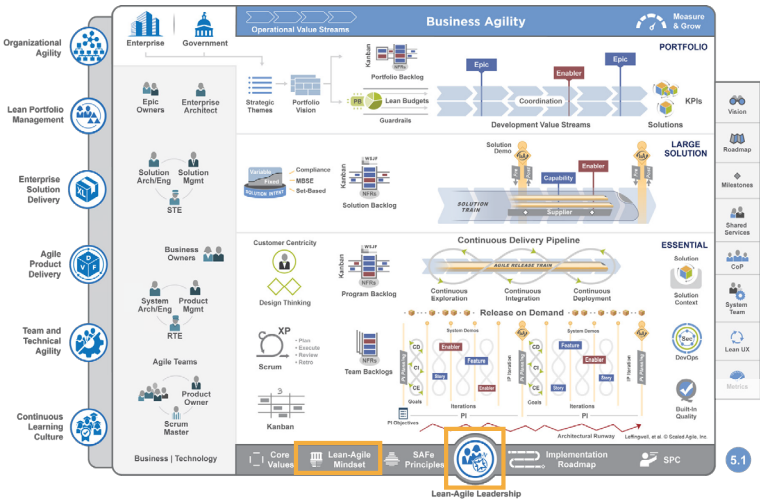
- Described what is necessary to thrive in the digital age
- Recognized SAFe as an operating system for Business Agility
- Explored the seven core competencies of Business Agility
- Explored the Lean-Agile Mindset
- Reviewed the SAFe Lean-Agile Principles
- Described Scrum, Kanban, and XP practices

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Lean-Agile Mindset" https://www.scaledagileframework.com/lean-agile-mindset/
- "Lean-Agile Leadership" https://www.scaledagileframework.com/lean-agile-leadership/
- "SAFe Lean-Agile Principles" https://www.scaledagileframework.com/safe-lean-agile-principles/



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Continue your SAFe journey with the following resources

Watch this three-minute video, *Navigating the Big Picture*, to understand how to use the SAFe Big Picture. **https://bit.ly/Video-NavigatingtheBigPicture**	Build your knowledge of the goals and methods of SAFe to achieve Business Agility with the "What is SAFe for Lean Enterprise" e-learning. **https://bit.ly/Community-GettingStarted**
Watch this four-minute video, *Lean-Agile Mindset,* to learn why a Lean-Agile mindset is an important Enabler for Business Agility. **https://bit.ly/Video-LeanAgileMindset**	Access the SAFe Collaborate template, "Experience Batch Sizes," to practice putting large and small batches through a system. **https://bit.ly/Template-BatchSize**
Download and share the "Introducing SAFe" toolkit to familiarize people in your organization with SAFe. **https://bit.ly/Community-ToolkitsandTemplates**	Complete the e-learning, "Agile Basics," to learn more about what Agile is and how it supports value delivery. **https://bit.ly/Community-GettingStarted**

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Lesson notes

Enter your notes below. If using a digital workbook, save your PDF often so you don't lose any of your notes.

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Lesson 2

Building an Agile Team

SAFe® Course - Attending this course gives students access to the SAFe® Practitioner exam and related preparation materials.

SP SAFe® 5

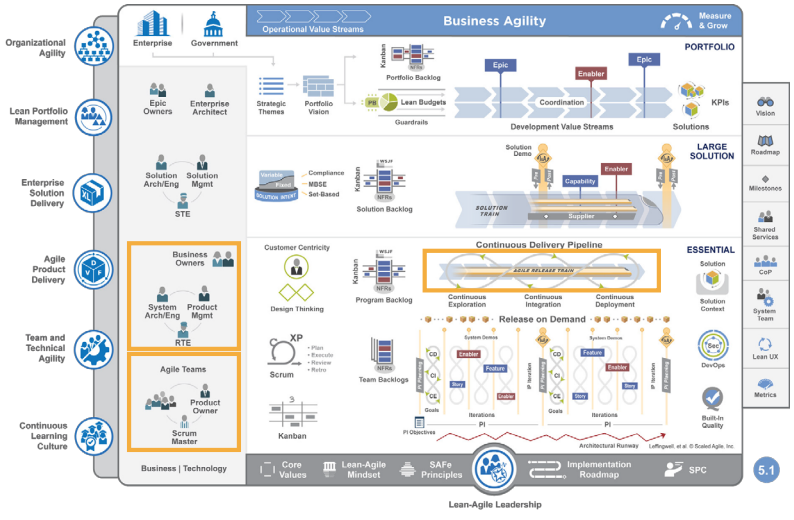
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Lesson Topics

2.1 Forming cross-functional Agile Teams

2.2 The Scrum Master and Product Owner roles

2.3 Organizing Agile Release Trains



2-2

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Learning objectives

At the end of this lesson, you should be able to:

- Build your Agile Team
- Explain the Scrum Master and Product Owner roles
- Identify the people and teams on the train, and learn about their roles

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2.1 Forming cross-functional Agile teams

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The power of a high performing team

We, the work, and the knowledge are all one.

- A self-organizing team dynamically interacts with itself and the organization
- Team members create new points of view and resolve contradictions through dialogue
- The team is energized with intentions, vision, interest, and mission
- Leaders provide autonomy, variety, trust, and commitment



2-5

Teams create and challenge norms

- There is creative chaos via demanding performance goals
- The team is challenged to question every norm of development
- Equal access to information at all levels is critical



2-6

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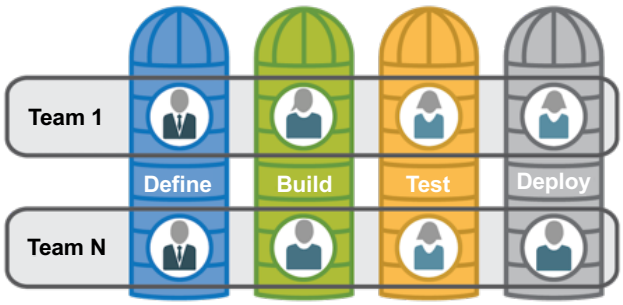
“It is amazing what you can accomplish if you do not care who gets the credit.”
—Harry S. Truman

2-7

Build cross-functional Agile Teams

Agile Teams are cross-functional, self-organizing entities that can define, build, test, and where applicable, deploy increments of value.

- Optimized for communication and delivery of value
- Deliver value every two weeks
- Contain two specialty roles:
  - Scrum Master
  - Product Owner



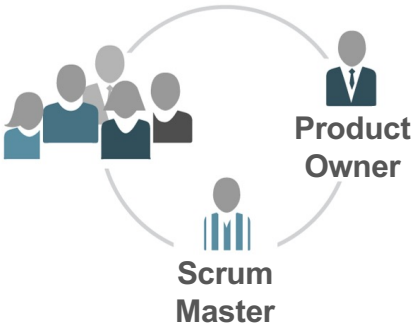
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Responsibilities of the Agile Team

- 5 to 11 team members
- Create and refine Stories and acceptance criteria
- Define, build, test, and deploy Stories
- Build quality in to each increment of the solution
- Develop and commit to team PI Objectives and Iteration goals



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Discussion: Agile Teams in your workplace

Prepare 5 min

Share 3 min

- **Step 1:** Discuss these questions
  - What would a cross-functional team look like at your workplace?
  - How would this change the results your team delivers?
- **Step 2:** Be prepared to share with the class



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